This week

SCHEDULE

Sample case: on LearnHub
- Read the case and think about the analyses
- We will carry out the lab analyses together and discuss the results.
- Quiz at the end of the lab session (should be present in the lab physically)

Lecture on Thursday:
- I will share a lecture video on LearnHub (due to minor health issues in the family).
- Listen to the video before the lab session.

Reading assignments:
- Chp 5.1-3: March 24
- Chp 5.4-8: April 07

AGENDA

Short review: Lab 4 & Class exercise

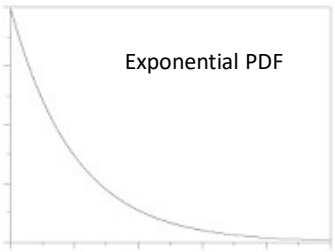
Confidence intervals
- Mean of normal random variables
- Mean of general random variables with a large sample size

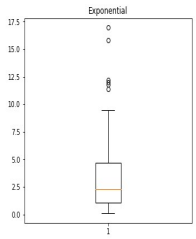
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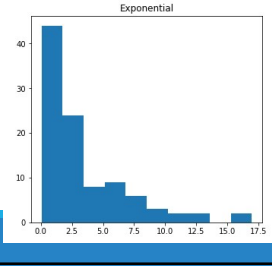
Lab 4 Exponential random variables

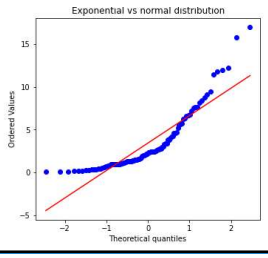
Exponential random variables with mean 4

Do the samples you generate confirm exponential(1/4)?









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Lab 4 Exponential random variables

Exponential random variables with mean 4

Do the samples you generate confirm exponential(1/4)?

Population

Mean	Std Dev	Min	Max	1st Q	Median	3rd Q	Skewness
4	4	0	∞	1.49	2.614	6.833	2

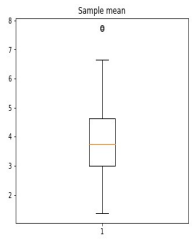
Sample

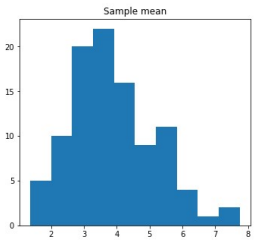
Mean	Std Dev	Min	Max	1st Q	Median	3rd Q	Skewness
3.8375	3.7477	0.0055	20.2145	1.5043	2.7290	4.8581	2.1329

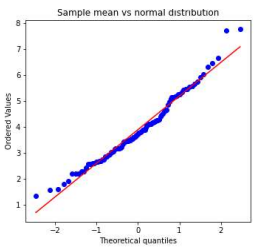
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Lab 4 Central Limit Theorem

Does the sample mean look like a normal random variable for n=10?







Population

Mean	Std Dev	Min	Max	1st Q	Median	3rd Q	Skewness
4	$4/\sqrt{10} = 1.265$	0	∞	--	--	--	<2

Sample

Mean	Std Dev	Min	Max	1st Q	Median	3rd Q	Skewness
4.0566	1.3548	1.4407	7.7347	3.0291	4.0444	4.8909	0.2231

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Lab 4 Central Limit Theorem

Does the sample mean look like a normal random variable for n=10?

Try different sample sizes. When does the sample mean behave like normal?

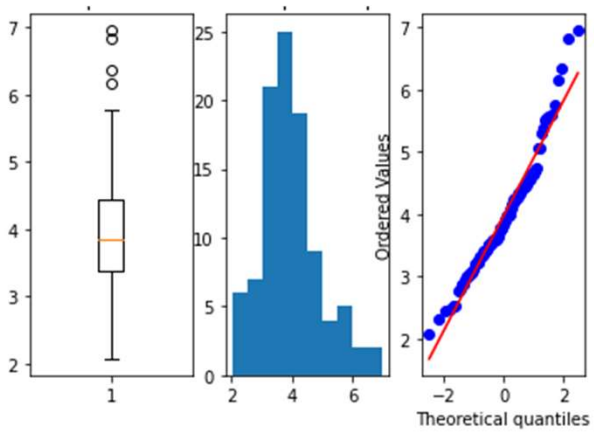
Can you verify the rule of thumb as n≥30?

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Lab 4 Central Limit Theorem

n=20

Sample mean plots

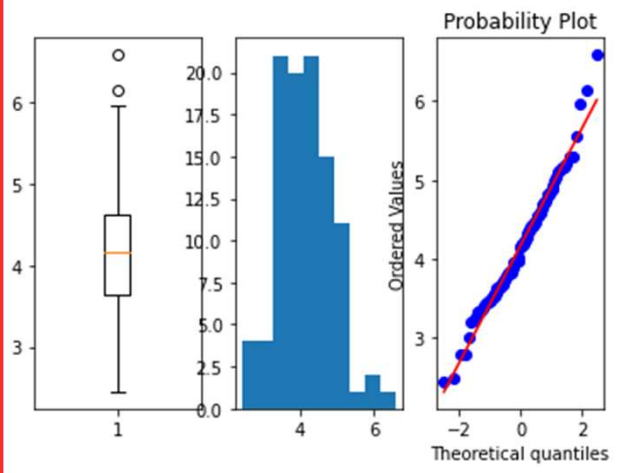


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Lab 4 Central Limit Theorem

n=30

Sample mean plots

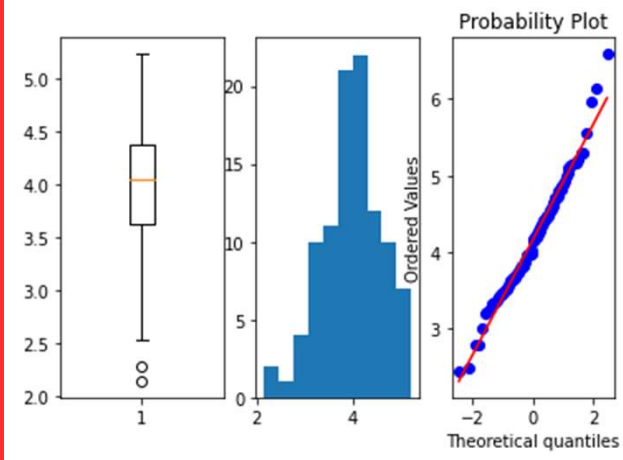


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Lab 4 Central Limit Theorem

n=40

Sample mean plots

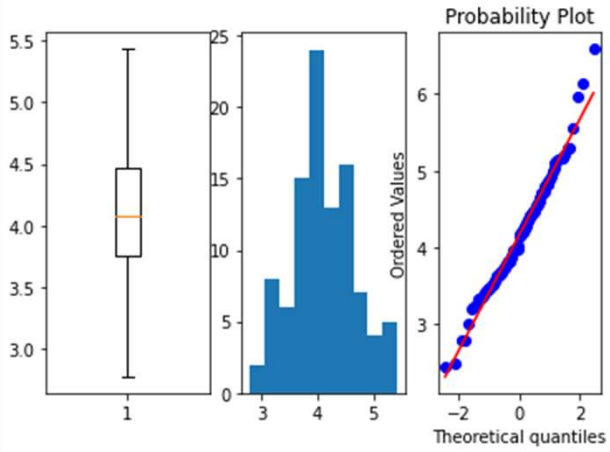


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Lab 4 Central Limit Theorem

n=50

Sample mean plots



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Lab 4 Central Limit Theorem

The meaning of CLT:

The sample mean behaves approximately normal, not the individual observations. The observations are still exponential.

$\{X_1, \ldots, X_n\}$: a random sample from an arbitrary distribution with $E[X_i]=\mu$ and $Var(X_i)=\sigma^2$

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \overset{approximately}{\sim} Normal\left(\mu, \frac{\sigma^2}{n}\right)$$

In lab 4: $\{X_i\}$ are exponential with rate ¼. Hence, $E[X_i]=\mu$ and $Var(X_i)=\mu^2$

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Lab 4 Confidence Intervals

$\{X_1, \ldots, X_n\}$: a normal random sample with $E[X_i]=5$ and $Var(X_i)=1$

The sample mean is also normal:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \sim Normal\left(\mu, \frac{\sigma^2}{n}\right)$$

The aim of confidence intervals is to find an interval estimator.

Assume that we know $Var(X_i)=1$.

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CI - Normal random sample

$X_1, \ldots, X_n$: random sample

$X_1, \ldots, X_n$ ~ Normal with mean μ and standard deviation σ.

Assume that σ is known.

A 100(1- α)% CI for the mean μ of a normal population when σ is known is given by:

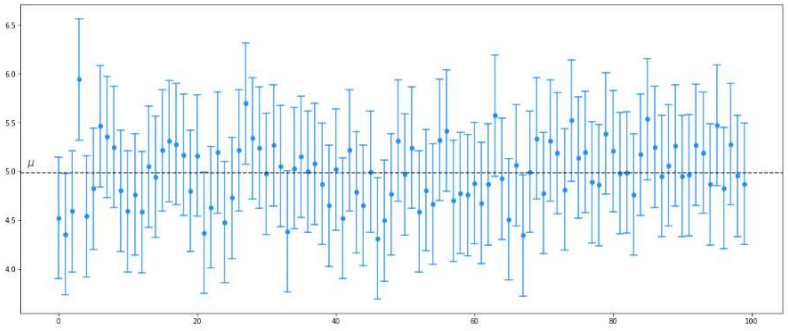
$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- Interpretation?
- Confidence Level, Precision & Choice of Sample Size?

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95% confidence intervals for 100 samples



The number of CIs that include the population mean 5 is an estimator for the confidence level 1-α.

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Lab 4 Confidence Intervals

Effect of α? For α =1%, $z_{0.01}$=2.33 and α =10%, $z_{0.1}$=1.28.

What happens to the precision? What happens to the confidence level?

Effect of the number of observations, n, when α is kept constant?

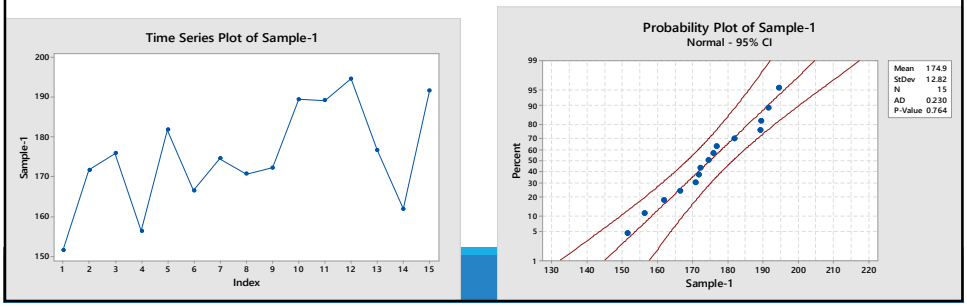
What happens to the precision? What happens to the confidence level?

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Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The population mean is 174.94.

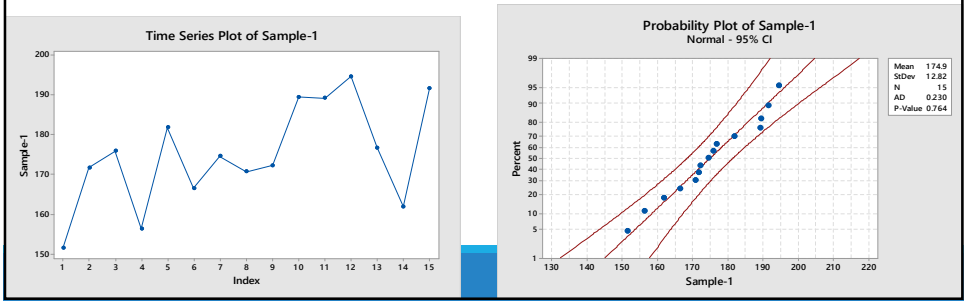


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Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The sample can be assumed to represent Turkey.

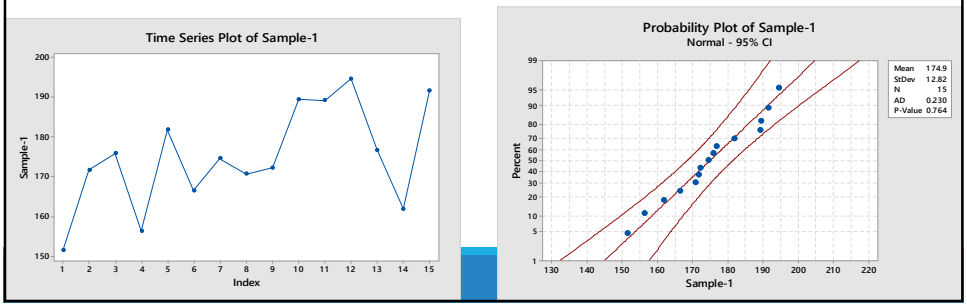


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Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The sample can be assumed random.

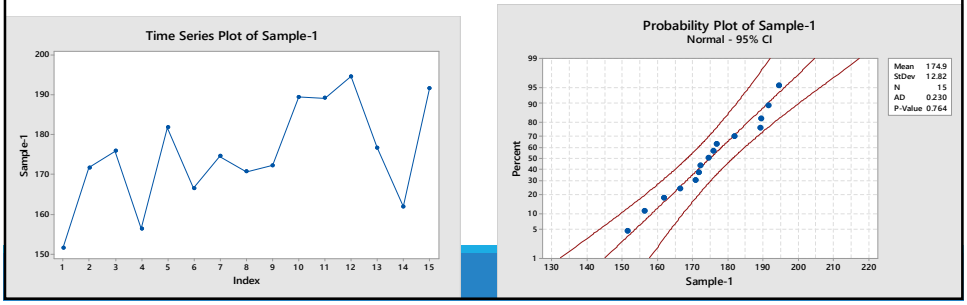


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Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The heights can be assumed normal.



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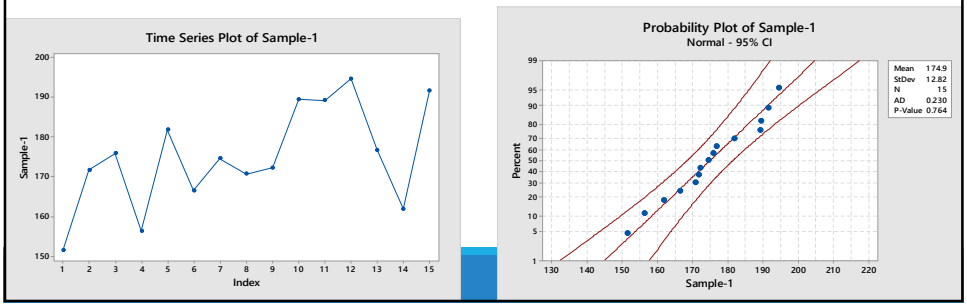
$z_{0.025}=1.96$

$z_{0.975}=-1.96$

Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The probability that the population mean is in (168, 182) is 95%.



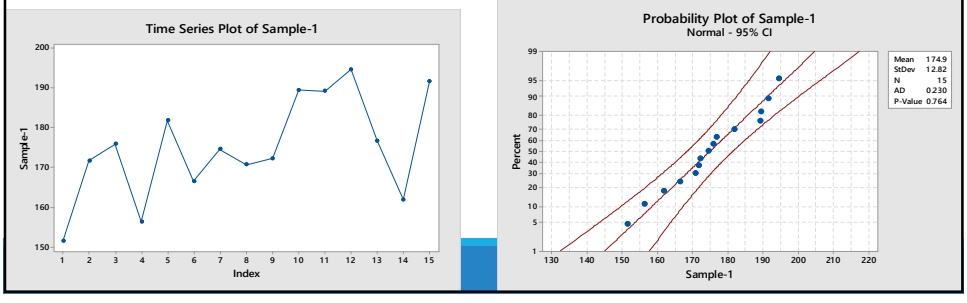
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$z_{0.025}=1.96$

Class exercise

Assume that you are interested in the mean height (μ) of all men in Turkey, where standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- The population mean is in (168, 182) with 95% confidence.



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Definition

A 100(1- α)% CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

σ =17
n=15
$\bar{x}$ =174.94
$z_{0.025}$=1.96

$$\left(174.94 - 1.96\frac{17}{\sqrt{15}}, 174.94 + 1.96\frac{17}{\sqrt{15}}\right) = (166.37, 183.54)$$

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Confidence Level, Precision & Choice of Sample Size

For fixed sample size: If the confidence level is increased, the width of CI is also increased so that the precision or accuracy (measured by the width of the CI) decreases if the reliability of CI is increased.

To have both high confidence & high precision, the only way is to increase the sample size.

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Width of the CI: $2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

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Precision & Choice of Sample Size

What is the width of 100(1- α)% CI? $2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

Let w be the desired width (so the precision you aim) of CI.

What is the required n? $n \geq \left(2z_{\alpha/2}\frac{\sigma}{w}\right)^2$

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Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ=2.4.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

If the desired width of 95%CI is 2, how many more delivery times should be recorded?

$\bar{x} = 6.05$

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To summarize:

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

$$100(1-\alpha)\%\text{CI}: \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- Interpretation?
- Precision, confidence, sample size?

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The same setting:

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can we define one-sided CIs?

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One-sided CIs

$P(L < \mu) = 1 - \alpha$, where L is a random variable.

means "probability that *L* is less than μ is 1-α".

Now assume that an observation is made on the random variable *L* so that *L=l*.

"$P(l < \mu) = 1 - \alpha$",
where l is an observation of the random variable L.

means "with 100(1-α)% confidence level μ is greater than l"

... and α is called the significance level.

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$100(1-\alpha)\%\text{CI}: \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$

One-sided CIs

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can you find a lower bound on the population mean μ with 1-α confidence?

Can you find an upper bound on the population mean μ with 1-α confidence?

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Confidence Interval for the Mean of a r.v. with a large sample

$X_1,\ldots,X_n$: random sample

Assumptions:

- n is very large. → $\bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- σ is known.

CI is the same as in normal r.v.s: $\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$

But σ is almost never known!!!

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Confidence Interval for the Mean of a r.v. with Unknown Variance:

$X_1,\ldots,X_n$: random sample (no specification on their distribution)

Assumptions:

- n is very large. → $\bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- both μ and σ are unknown.

↓

σ will be estimated by sample standard deviation S.

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The change in C. I.?

$\frac{\bar{X}-\mu}{S/\sqrt{n}}$ ⟶ more variable compared to $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$

but still OK, since n is large!!!

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Proposition

If n is sufficiently large:

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \overset{approx.}{\sim} N(0,1)$$

so that

$$\left(\bar{x} - z_{\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}\right)$$

is a large-sample interval for μ with confidence level approximately 100(1-α)%. This formula is valid regardless of the population distribution.

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Rule of thumb

n>40 is generally sufficient for all kinds of random variables. To "fine-tune" this rule for specific probability distributions, you can run simulations.

More conservative than CLT because of the additional variability introduced by using S instead of σ.

You can experiment with "n>40."

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A Large-Sample CI for a Population Proportion

X : # of successes in a random sample with n << population size

p : proportion of 'success' in a population

X~bin(n,p)

For large n with $np \geq 10$ and $n(1-p) \geq 10$, X is approximately normal.

$$E(X) = np$$
$$V(X) = np(1-p)$$

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A Large-Sample CI for a Population Proportion

X~bin(n,p) $E(X) = np$
$V(X) = np(1-p)$

For large n with np ≥ 5 and n(1-p) ≥ 5, X is approximately normal.

Given this what is the distribution of $\hat{P} = \frac{X}{n}$?

Parameters?

Now find the corresponding confidence interval for p using this information.

Estimate the variance of X if necessary.

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Estimate standard deviation of p

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad \rightarrow \quad \hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

An approximate confidence interval for a population proportion p with confidence level approximately 100(1-α)% is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

which is a traditional CI used for decades. This always needs:

$$n\hat{p} \geq 10 \text{ and } n(1-\hat{p}) \geq 10$$

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Example

100 architecture students in different universities in turkey are sampled, and it is found that 60 of them are female and 40 are male. Construct an approximate 95% CI for p, where p is the probability of having a female architecture student.

$$0.6 \pm 1.96\sqrt{\frac{0.6 \times 0.4}{100}} = (0.5040, 0.6960)$$

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Estimate standard deviation of p

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad \rightarrow \quad \hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

An approximate confidence interval for a population proportion p with confidence level approximately 100(1-α)% is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

which is a traditional CI used for decades. This always needs:

$$n\hat{p} \geq 10 \text{ and } n(1-\hat{p}) \geq 10$$

However, the intervals may cover the true proportion p in less than 100(1-α)% of the intervals.

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New research:

Recent research shows that a slight modification of *n* and the following estimate of *p* provide a good confidence interval:

Define $\tilde{n} = n + 4$ and $\tilde{p} = \frac{X+2}{\tilde{n}}$. → Add 2 observations on both outcomes, which increases the uncertainty, so increases the width of the CI.

An approximate confidence interval for a population proportion p with confidence level approximately 100(1-α)% is:

$$\tilde{p} \pm z_{\alpha/2}\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}.$$

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Example

$\tilde{n} = n + 4 = 104$

$\tilde{p} = \frac{X+2}{\tilde{n}} = \frac{60+2}{100+4} = 0.596$

100 architecture students in different universities in turkey are sampled, and it is found that 60 of them are female and 40 are male. Construct an approximate 95% CI for p, where p is the probability of having a female architecture student.

Width of the CIs:

Traditional CI: $0.6 \pm 1.96\sqrt{\frac{0.6\times0.4}{100}} = (0.5040, 0.6960)$ 0.192039996

New CI: $0.596 \pm 1.96\sqrt{\frac{0.59\ \times0.404}{104}} = (0.4998, 0.6962)$ 0.196353405

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Inferences based on two samples

$X_1,\ldots,X_m$: a random sample from a population with $\mu_1 \& \sigma_1^2$

$Y_1,\ldots, Y_n$: a random sample from a population with $\mu_2 \& \sigma_2^2$

X & Y samples are independent of one another.

Can we find a confidence interval around μ_1 - μ_2?

What can you say about $\bar{X} - \bar{Y}$?

$$E(\bar{X} - \bar{Y}) = ?$$
$$V(\bar{X} - \bar{Y}) = ?$$

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A Confidence Interval for the Difference Between Two Means

When both samples are normal and the values of σ_X and σ_Y are known :

If the two samples are independent, then a level $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}.$$

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A Confidence Interval for the Difference Between Two Means

When the two samples are from an arbitrary distribution, the values of σ_X *and* σ_Y *are unknown and both* n_X *and* n_Y *are large:*

If the two samples are independent, then a level $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}.$$

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